Developing a Multi-Microphone Impedancetube Focused on Determining Room Acoustic Material Properties at Low Frequencies

Bachelor Project ESD6

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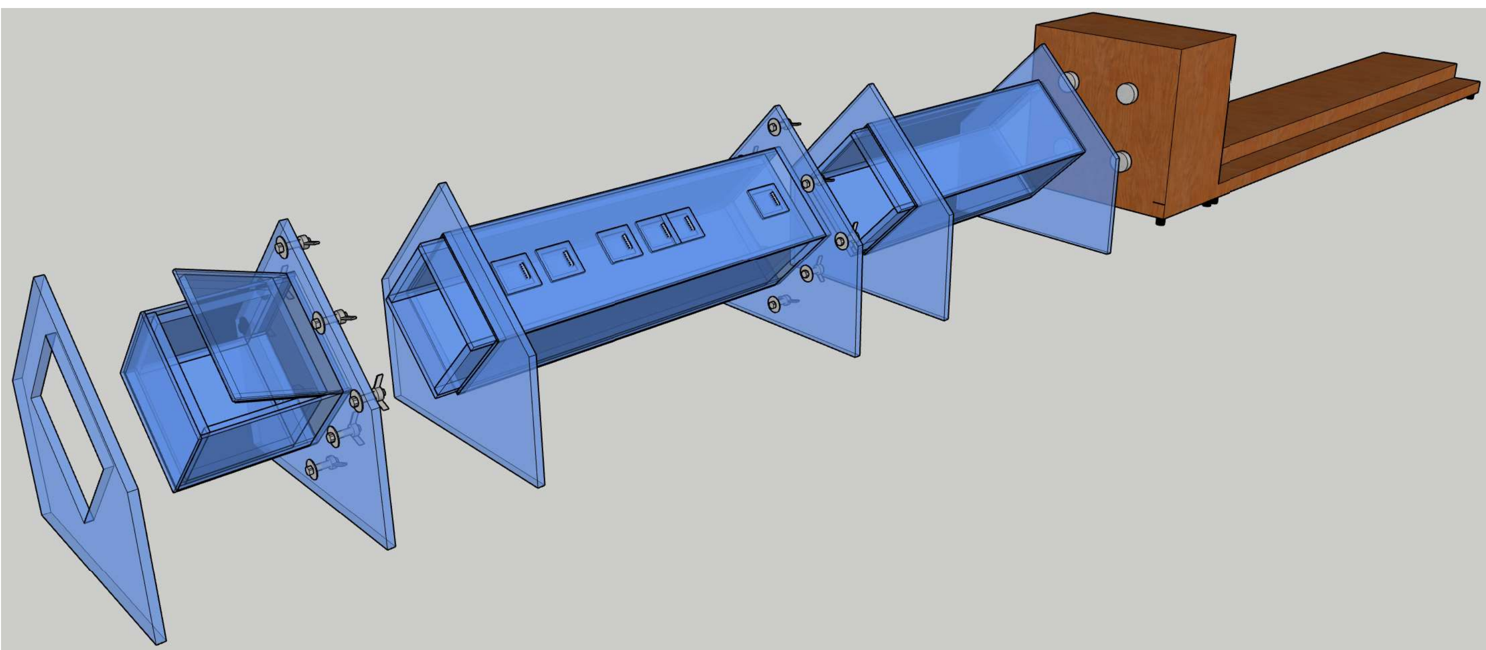

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AALBORG UNIVERSITY
STUDENT REPORT

Title:
Developing a Multi-Microphone Impedancetube Focused on Determining Room Acoustic Material Properties at Low Frequencies

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Abstract:

Good acoustics is an ever-growing subject within modern construction through various techniques using known methods and simulating new ideas. A common issue across the industry is treatment against low-frequency noise, as most test equipment is not operational below 100Hz. This project addresses the gap in accurately characterizing the low-frequency properties of construction materials using impedance tubes. The project exemplifies how an impedance tube fit for low-frequency analysis can be developed physically and electronically. The system proposal utilises a Teensy 4.1 microcontroller with an audio shield, and some highly customized signal processing techniques to create maximum length sequence signals and derive impulse responses in both the frequency and time domains directly from the microcontroller. Although the end result hasn't been able to successfully derive reflection and absorption coefficient yet, the project as a whole has succeeded in developing an impedance tube meeting DS/ISO and ASTM standards, and has made significant headway in complete acoustic analysis on a cheap microcontroller.